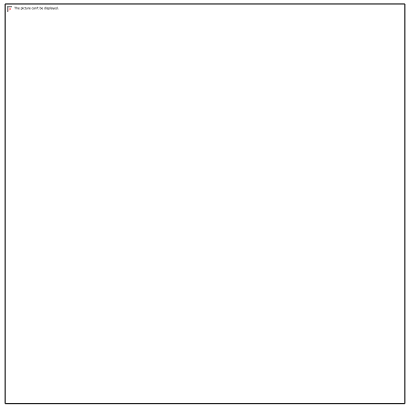
Toward a Standard Benchmark Format and Suite for Floating-Point Analysis



Nasrine Damouche, Matthieu Martel, **Pavel Panchekha**,
Chen Qiu, Alexander Sanchez-Stern, Zachary Tatlock.

Incredible progress…

Automatic Verification
Fluctuat [SAS'13]
Rosa [POPL'14]
FPTaylor [FM'15]

Optimization
STOKE [PLDI'14]

Improvement
Salsa [FMICS'15]
Herbie [PLDI'15]

Mechanized Proofs
Wave equation [ITP'10]
Rounding error [NSV'16]

Rapid improvement in hard problems!

Incredible progress...

We want our community
to keep progressing!

Next
???

Rapid improvement in hard problems!

We want our community
to keep progressing!

As community grows,
growing pains appear



FPBench

FPBench is **community infrastructure** for **cooperation** and **comparison** in the FP community.

http://fpbench.org



FPBench

β

Common format

Benchmark suite

Named measures

$$\sqrt{x+1} - \sqrt{x}$$

```
(FPCore (x)
  :name “Sqrt Difference”
  :cite (hamming-87)
  :pre (> x 0)
  (- (sqrt (+ x 1)) (sqrt x)))
```

Metadata

Preconditions

```
(FPCore (x0)
  :name “Sine Newton”
  :cite (darulova-kuncak-2014)
  :pre (< (abs x0) 1)
  (while (< i 10)
    ([i 0 (+ i 1)]
     [x x0 (+ (+ (- x (/ (pow x 3) 6.0))
                 (/ (pow x 5) 120.0))
                 (/ (pow x 7) 5040.0))
              (+ (+ (- 1.0 (/ (* x x) 2.0))
                 (/ (pow x 4) 24.0))
                 (/ (pow x 6) 720.0))))])
    x))
```

Loops

Common functions

FPCore common format

Simple to use

S-expression syntax
Purely functional
No control flow analysis

Expressive

All C, Fortran functions
Loops, conditionals
Tools support parts

Extensible

Metadata properties
Tool-specific metadata
Input or output format

Generate from higher-level, imperative FPImp lang.



FPBench

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Common format

Benchmark suite

Named measures

Simple to implement

Covers all existing uses

Simple to extend, specialize



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FPBench benchmark suite

72 total benchmarks

Drawn from existing papers

Annotated with source, ranges, description, citation

FPBench benchmark suite

Existing programs		Rich features		Diverse domains	
FPTaylor	29	Arith	72	Textbook	59
Herbie	28	Expt	16	Math Alg	6
Rosa	6	Trig	11	Emb Sys	4
Salsa	9	Loop	12	Sci Comp	3
		Branch	3		



FPBench

β

Common format

Simple to implement

Covers all existing uses

Simple to extend, specialize

Benchmark suite

From existing projects

Cover many domains

Grows over time

Named measures



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Simple to implement

Covers all existing uses

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From existing projects

Cover many domains

Grows over time

FPBench measures

Formal definitions of accuracy measures

Described along 5 axes

Standard measures so tools agree

FPBench axes of measurement

Scaling vs. non-scaling	Absolute, relative, ULPs, bits, …
Forward vs. backward	Fixed input error vs fixed output error
Maximum vs. average	vs
Sound vs. statistical	Formal guarantees vs mathematical accuracy
Improvement	



FPBench

β

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Benchmark suite

From existing projects

Cover many domains

Grows over time

Named measures

Terms for measuring error

Standard across tools

Flexible but rigorous



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- Common format
- Benchmark suite
- Named measures

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